

Assembly memory map

This is the authoritative map of every absolute-origin block in Public Beta 0.8. Addresses in the **End** column are inclusive. The build does not trust `AORG` by itself: `build.ps1` reads the start and end labels from the assembler listings and fails if a block moves, grows into the next block, or if an unreviewed `AORG` is added.

The same measured regions are written to `build/manifest.json` after every successful build. That generated copy uses end-exclusive addresses so its byte counts can be checked directly.

Direct-mapped 16 KiB image

CPU address	Size	Owner
>6000->7FFF	8 KiB	Interactive extension ROM
>C000->CFFF	4 KiB	Tester U7 SRAM workspace and diagnostic state
>D000->D537	1,336 bytes	Relocated raster-effect runtime in tester U7 SRAM
>D538->DFFF	2,760 bytes	Unused upper U7 SRAM
>E000->FFFF	8 KiB	Automatic/fail-safe core ROM

The W27C512 programmer image stores the extension at physical file offset `>6000`, the core at `>E000`, and `>FF` everywhere else.

Fixed core: `src/HEXDIAG.a99`

Start	End	Used	Free before next block	Contents
>E000	>F0AD	4,270	82	Header, automatic tests, services, text, and private 96-character font
>F100	>F229	298	0	Exact MegaDemo PSG phrase
>F22A	>F3F1	456	340	Enhanced-VDP detector and shared extension strings
>F546	>F861	796	20	VDP assets and keyboard-matrix helpers
>F876	>FDAD	1,336	0	Byte-exact relocated U7 runtime image
>FDAE	>FDF6	73	1	GROM-result formatter and labels
>FDF8	>FDFB	4	4	Extension ABI marker and version
>FE00	>FEF7	248	8	Extension service gateway
>FF00	>FF19	26	222	LOAD entry
>FFF8	>FFFF	8	0	Image marker, checksum word, workspace and LOAD vectors

The private ASCII 32-127 font currently occupies `>EDAE->F0AD` inside the first region. It is checked separately as exactly 768 bytes. This guard exists because a former backward `AORG >F070` silently replaced the glyphs for lower-case `x`, `y`, `z`, `{`, `|`, `}`, `~`, and character 127.

Extension: `src/HEXEXT.a99`

Start	End	Used	Free before next block	Contents
<code>>6000</code>	<code>>7EEF</code>	7,920	16	Header, menus, interactive code, strings, and data
<code>>7F00</code>	<code>>7F2F</code>	48	0	TI-99/4A/QI keyboard matrix table
<code>>7F30</code>	<code>>7F57</code>	40	0	TI-99/4 keyboard matrix table
<code>>7F58</code>	<code>>7F7F</code>	40	0	TI-99/4 SHIFT/function translation table
<code>>7F80</code>	<code>>7FBF</code>	64	62	TI-99/4A SHIFT/function translation table
<code>>7FFE</code>	<code>>7FFF</code>	2	0	Extension checksum correction word

The build requires the variable extension to end no later than `>7EEF`, preserving at least 16 bytes of deliberate headroom before `>7F00`. The current build ends at `>7EEF`, leaving 16 bytes. This guard exists because variable text once reached `>7F00` and silently replaced the first keyboard-table byte and a string terminator.

Relocated U7 image: `src/MEGAU7.a99`

`MEGAU7.a99` is relocatable assembler output with `XORG >D000`, rather than a ROM `AORG`. Its listing must report `MSTART=>D000` and `MEND=>D538`, exactly 1,336 bytes. The core embeds that exact binary at `>F876->FDAD`; both lengths are checked, so the runtime and embedded copy cannot drift apart.

Enforced origin inventory

The core contains ten reviewed `AORG` directives:

```
>E000 >F100 >F22A >F546 >F876 >FDAE >FDF8 >FE00 >FF00 >FFF8
```

The extension contains five:

```
>6000 >7F00 >7F30 >7F80 >7FFE
```

The checker rejects a changed count, changed address, repeated address, or backward address before it evaluates region sizes. This makes adding a new fixed block an explicit map change instead of an invisible overwrite.